Date. No.

Lesson 8

1. Embedding bu avvalo tokenshning sonli vektorlarga aylantirilishi

Text → tokenizer → Embedding →
↳ Token IDs
Transformer

Embedding nega kerak sabab Transformer-dagi Attention layer integerlardan o'rgana olmaydi. Yani Token IDs bersak hechnarsa o'rgana olmaydi.

Embeding jarayonining 2ta muhim qismi bor:

1. WTE → Word token Embedding
2. WPE → Word Position Embedding

WTE da wordni tokenga aylantirib keyin ID berib shu ID ni Embedding qilib vectorga aylantirib oladi.

Ex: "cat" - 876 → [0.2, -0.8, ...]
vektor ölchami nechchidir berishimizga bog'liq

WPE esa bu usul bilan modelga token qayerda joylashganini bildiramiz.

Men	döstimni	uchratdim.
pos 1	pos-2	pos-3
Döstim	meni	uchratdi
pos-1	pos-2	pos-3

Bu yerda Men 1-gapda 1-Positionda 2-gapda 2-positionda. Pozitsiyasiga qarab model örganadi.

2ta usuldagi embedding birlashib modelga yani keyingi qatlam Transformerga (NN) uzatiladi.

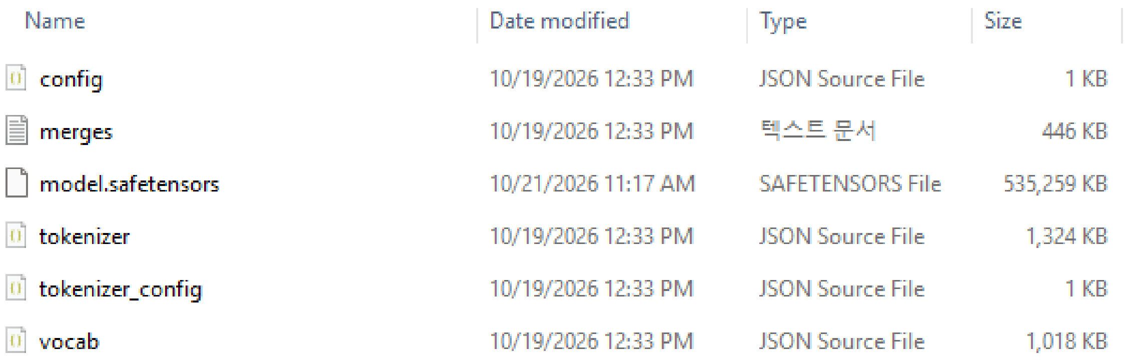
Basic questions

1. Pre-Norm va Post-Norm farqi
2. SwiGLU va GeGLU nega hozir foydalanilyapdi sabab.
3. RoPE nega kōp hollarda yaxshi
4. Vocabulary vs tokenization
5. Weight decay vazifasi va ishlashi.
6. Gradient clipping, Z-loss va QK-Norm nima? va qachon qanday holda ishlatamiz.
7. GQA KV cache memoryni qanday save qiladi.
8. WTE vs WPE ularni vazifalari va tarifi.
9. Nega Transformation da kōpincha Pre-Normni ishlatamiz.
10. Nega ReLU va GeLU eskirdi.

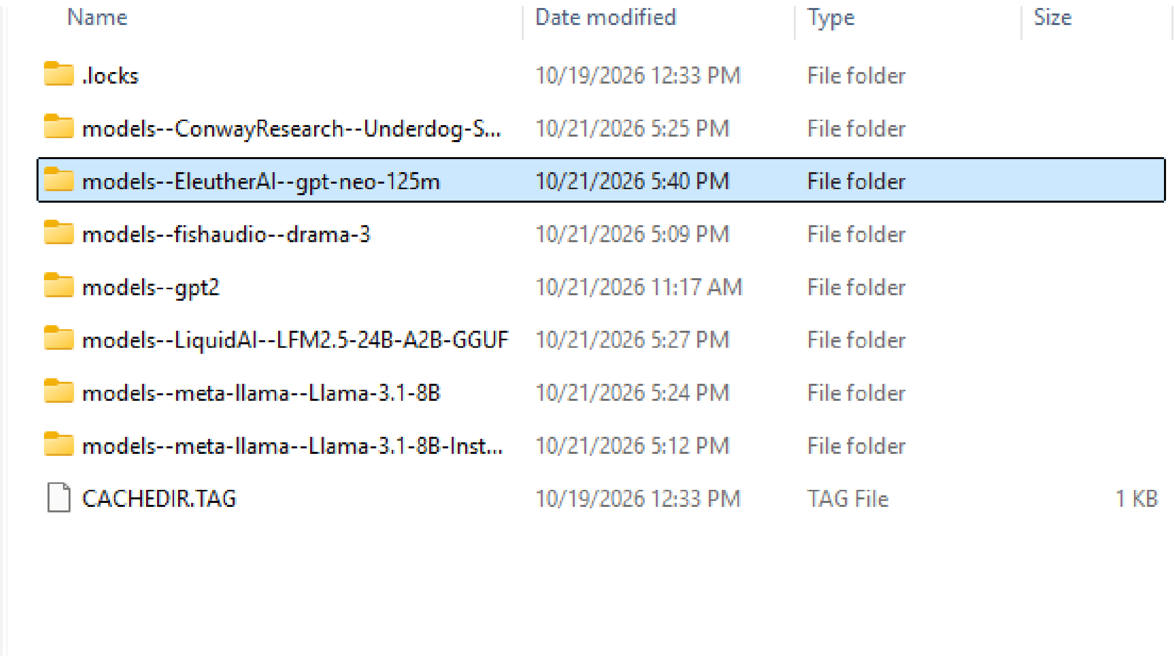
follow-up

1. faqat WTE ~~yok~~ ishlatsak nima bōladi.
2. GQA va MHA nimasi bilan farqli bir biridan yani + va - taraflari
3. Gradient Clipping qanday qilib trainingda instability ga yordam beradi.
4. Weight decay juda yuqori bōlsa nima sodir bōlishi m-n

   supervised
5. SML va LLM da hyperparameters farqi va ishlashi.
6. Agar Transformerga kirishidan oldin input Norm ishlatilinmasa ?
7. SwiGLU ReLU dan bitta afzalligi bor nima ?
8. Dropout nega => 0 ?
9. Ōzi nega frontier models 1 xil arxitekturadan foydalanib qōymaydi ?
10. ~~Nega~~ Model parameters kōp va kam holatlarda nima bōlishi m-n.



Name	Date modified	Type	Size
config	10/19/2026 12:33 PM	JSON Source File	1 KB
merges	10/19/2026 12:33 PM	텍스트 문서	446 KB
model.safetensors	10/21/2026 11:17 AM	SAFETENSORS File	535,259 KB
tokenizer	10/19/2026 12:33 PM	JSON Source File	1,324 KB
tokenizer_config	10/19/2026 12:33 PM	JSON Source File	1 KB
vocab	10/19/2026 12:33 PM	JSON Source File	1,018 KB



Name	Date modified	Type	Size
.locks	10/19/2026 12:33 PM	File folder	
models--ConwayResearch--Underdog-S...	10/21/2026 5:25 PM	File folder	
models--EleutherAI--gpt-neo-125m	10/21/2026 5:40 PM	File folder	
models--fishaudio--drama-3	10/21/2026 5:09 PM	File folder	
models--gpt2	10/21/2026 11:17 AM	File folder	
models--LiquidAI--LFM2.5-24B-A2B-GGUF	10/21/2026 5:27 PM	File folder	
models--meta-llama--Llama-3.1-8B	10/21/2026 5:24 PM	File folder	
models--meta-llama--Llama-3.1-8B-Inst...	10/21/2026 5:12 PM	File folder	
CACHEDIR.TAG	10/19/2026 12:33 PM	TAG File	1 KB